

Chengwen QU

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Canadian citizen

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[Tech blog](#) | [Github](#)

SUMMARY

I am a software engineer with a unique background spanning engineering and modern software development. Experienced in object-oriented design, cloud native solution engineering, database schema design, I built data-intensive enterprise system, empowering thousands of organizations. My work has consistently translated into real world impacts, with a proven track record of delivering solutions from proof of concepts through to production.

EDUCATION

Memorial University of Newfoundland (2013 - 2018)

Bachelor of Engineering, Ocean and Naval Architectural Engineering

Algonquin College (2020 - 2021)

Associate degree (honours),
Computer Programming

Georgia Institute of Technology
(2022- present, part time online)

Master of Computer Science

Additional Info

Awards/Training: AWS Developing Serverless Solutions, AWS DevOps Engineering, Nvidia Fundamentals of Deep Learning, Nvidia Building AI-Based Cybersecurity Pipelines

Skills: C, Java, Python, SQL, Object-Oriented Design, Data Storage Design, Test-driven development, Git, Bash Scripting, Gradle, Docker, Apache Kafka, Postgres, DynamoDB, Redis, SQS, AWS Step Function, Terraform, Kubernetes

WORK EXPERIENCE

Trend Micro Canada Technologies (2021 – present)

Senior Software Developer (2023.2-present)

- Performed systems analysis, including requirement clarification, solution design, and impact assessment for system changes
- Maintained and developed new features for the Compliance Risk Exposure Management app for TrendAI Vision One Platform
- Designed and architected validator service in the new alerting microservice
- Optimized Java code performance through caching strategies and efficient data processing techniques
- Troubleshooted issues and provide solutions related to forwarding of SIEM events for customers
- Improved CI/CD pipelines, testing, and deployment processes to ensure high-quality releases
- Mentored junior developers and co-op student

Backend Software Developer (2021.11- 2023.2)

- Developed and maintained data-pipeline services enabling near real-time data integration across multiple databases and storage systems, using Apache Kafka
- Provisioned and optimized infra
- Wrote non-application code, such as infrastructure as code, pipeline as code, unit tests, metrics/alerts, monitoring etc.

TechnipFMC Canada (2018 – 2020)

Project Engineer

- Performed engineering work related to design, procurement, manufacturing and installation of offshore oil production structures
- Wrote offshore installation procedures detailing installation methods, sequence, required equipment, installation criteria
- Offshore execution of a work scope

Notable Personal Projects

[Stock analysis and trading website](#) (2020)

- A virtual stock trading website where users can browse 5000+ stocks and their real-time price, buy and sell stocks

[Auctions](#) (2021)

- An auction website where user can post and bid for items, with some built-in social media functionalities

[Android App](#) (2021)

- pick dates and browse 10,000+ NASA images and its feature, vote Favorites

[Software defined network](#) (2023)

- Router design and algorithm, BGP hijacking

[Multi-threaded http library – System programming](#) (2024)

- Multi-threaded http-based server client library written in C

[Data Visualization App](#)

- Visualizing Canada federal election results

[Pixel Studio Website](#)

- website design for a local small business